

# Implementation Identity Matters: Post-Route Trade-offs in Memory ECC Hardware

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**Abstract**—Error-correcting-code (ECC) capability alone does not determine whether a protection scheme is practical in hardware. Algorithm-level comparisons omit microarchitectural latency and the area, timing, routing, and energy effects introduced during physical implementation. We present an identity-preserving evaluation method that admits correctness-qualified ECC implementations, separates guaranteed correction from exhaustive out-of-guarantee observations, and characterizes accepted RTL under one pinned post-route flow. In SKY130HD, two temporally aligned exact-equivalent SECDED (72,64) implementations occupy non-dominated points: relative to the combinational design, pipelining increases standard-cell area by 36.702% and adds two cycles of request latency, but raises achieved Fmax by 45.911% and reduces steady-stream energy per operation by 24.016%. The evaluated shortened BCH (78,64, $t=2$ ) implementation corrects all weight-2 errors but uses 207.356% more area than combinational SECDED and misses the common 10 ns target. These results are specific to the evaluated RTL, technology, corner, and physical policy. They show why reliable selection must preserve implementation identity, timing feasibility, and missing evidence rather than infer physical quality from coding strength.

**Index Terms**—error-correcting codes, memory reliability, ECC hardware, physical design, cross-layer design, energy efficiency

## I. INTRODUCTION

Error-correcting codes (ECCs) are fundamental to protecting memory systems from data corruption. Memory ECCs are commonly differentiated by correction and detection capability, redundancy, and decoding complexity. Those properties establish what a code can protect, but implementation decisions determine whether that protection is practical within area, clock, latency, routing, and energy constraints.

Algorithm-level descriptions do not expose standard-cell area, achievable timing after routing, interconnect burden, or activity-dependent energy. Moreover, the mapping from a code to hardware is not unique: two architectures can implement exactly the same coding semantics while inserting different register boundaries and logic structures. Hardware-aware SECDED construction and checker optimization are well established [1]–[4], as are cross-layer studies that trade ECC strength against memory cost [5], [6]. This makes a controlled implementation comparison more useful than a claim that physical cost has been overlooked.

We use the workflow in Fig. 1: correctness and implementation identity are qualified before physical comparison; guaranteed reliability semantics remain separate from observations outside that guarantee; accepted designs are implemented under common constraints; and only then are reliability and physical cost interpreted together. The procedure is fail-closed:

unsupported or incomplete measurements remain unavailable rather than being imputed.

The strongest controlled contrast is between combinational and pipelined extended-Hamming SECDED (72,64) RTL. Exact equivalence after temporal alignment fixes their coding semantics. Pipelining increases area by 36.702% and detailed wirelength by 12.067%, and adds two cycles of request latency, while increasing achieved Fmax by 45.911% and reducing equal-work, steady-stream energy per operation by 24.016%. A second contrast is the evaluated BCH (78,64, $t=2$ ) RTL: relative to combinational SECDED, it uses 207.356% more area and 310.975% more wirelength, achieves 72.258% lower Fmax, and fails the common 10 ns target with -4.78052 ns worst negative slack (WNS).

This paper makes three contributions:

- 1) an identity-preserving cross-layer evaluation method that admits correctness-qualified ECC implementations and keeps guarantees, observations, architecture, and unavailable physical data distinct;
- 2) a quantified non-dominated post-route trade-off between two exact-equivalent SECDED microarchitectures; and
- 3) an implementation-scoped correction-versus-feasibility result for the evaluated BCH RTL, while retaining unavailable Hsiao PPA explicitly rather than estimating it.

All findings are bounded to the evaluated RTL, SKY130HD environment, and frozen experimental conditions.

## II. BACKGROUND AND RELATED WORK

### A. ECC Algorithms and Hardware Implementations

Extended-Hamming SECDED corrects one error and detects two, while odd-weight-column Hsiao codes reduce checker complexity for the same guarantee [1], [7]. Ghosh *et al.* used parity-check-matrix freedom to reduce Hamming/Hsiao checker power while monitoring area and delay [2], [3]; Reviriego *et al.* constructed SEC/SECDED codes to reduce encoder/decoder delay and area [4]. Naseer and Draper compared SEC, Hsiao SECDED, double-error-correcting (DEC), and DEC-TED parallel hardware for SRAM multi-bit upsets [8]. These works establish both hardware-conscious code design and correction-cost comparison.

BCH hardware has its own broad architecture space. Chien-search power can be reduced through RTL-level activity management [9], and recent conventional/direct decoder comparisons show large architecture-dependent area, throughput, and latency differences for common BCH parameters [10].

Accordingly, our BCH measurement is one implementation point, not a bound on BCH as a family.

### B. Reliability-Aware Memory Protection

System-level work jointly evaluates reliability and memory overhead. Virtualized ECC and LOT-ECC trade correction strength against storage, traffic, performance, and energy [11], [12]; Bamboo ECC changes code organization to improve protection under channel constraints [13]. FaultSim provides field-model-driven simulation of SECDED/Chipkill-class resilience [14]. Those studies are broader in memory organization and operational reliability than ours. We instead report finite canonical-coordinate guarantees/observations and do not claim FIT, SER, or field-weighted rates.

### C. Hardware-Cost and Cross-Layer Evaluation

DATE studies of STT-MRAM explicitly co-design correction strength with device, cell, array, area, energy, and yield considerations [5], [6]. The present work therefore does not claim cross-layer ECC/cost analysis itself as new. Its narrower focus is a controlled post-route study: correctness-qualified RTL identities, two exact-equivalent SECDED microarchitectures, a shared physical methodology, equal useful work for energy, and explicit missing/infeasible points. OpenROAD supplies the open digital-flow foundation [15]; our contribution is the ECC experiment and evidence discipline built on a pinned instance of that flow.

## III. CROSS-LAYER ECC EVALUATION METHODOLOGY

### A. Correctness and Identity Qualification

Each candidate binds a code identity to a specific RTL architecture and interface. Only candidates with sufficient functional evidence enter comparison. Qualification records the payload/codeword relation, correction and detection guarantee, latency, initiation interval (II), and the correspondence between the tested implementation and the physically characterized RTL. For the two conventional SECDED candidates, exhaustive formal equivalence after a two-cycle temporal alignment establishes identical extended-Hamming (72,64) input/output semantics. Thus, their physical difference is a microarchitectural treatment rather than a different code.

This step prevents nominal labels such as “SECDED” or “BCH” from silently pooling measurements from unrelated implementations. It also prevents a physically successful but functionally unqualified design from becoming a comparison point.

The comparison contract fixes the transaction boundary as well as the code. Encoder and decoder channels are independent, parallel interfaces with no backpressure, and each can accept one valid transaction per cycle. Every implementation carries the same 64 useful payload bits, while the realized 72- or 78-bit codeword width is retained as an implementation cost. Latency is measured from accepted request to corresponding response; II describes successive acceptance. Consequently, a three-cycle design with II=1 is neither mislabeled as lower-throughput nor treated as equal-latency to a one-cycle design.

Qualification and characterization are deliberately separated. Correctness evidence determines admission; it cannot be rescued by favorable PPA. Conversely, a candidate that lacks physical evidence is not assigned a synthetic worst-case point. This distinction is important for Pareto analysis: an unavailable coordinate is epistemically different from a measured but unattractive coordinate.

### B. Reliability Semantics

We separate guarantees from observations. The conventional and Hsiao SECDED implementations correct all single-coordinate errors and detect all double-coordinate errors. The shortened BCH (78,64, $t=2$ ) implementation corrects all one- and two-coordinate errors in its proven universe.

For context, we also exhaustively enumerate weight-3 masks over canonical codeword coordinates and classify decoded outcomes as silent data corruption (SDC) or detected uncorrectable error (DUE). These are finite-universe observations beyond the guaranteed correction radius. They are not FIT, SER, operational probabilities, or field-weighted rates. In particular, BCH spans  $\binom{78}{3}$  masks while SECDED spans  $\binom{72}{3}$ ; fractions must not be treated as samples from one common physical fault distribution.

### C. Common Physical Characterization

Accepted RTL enters one pinned OpenROAD-flow-scripts (ORFS) environment. Technology, corner, target clock, I/O delays, output load, utilization policy, placement density, seed, and worker count are common. The area metric is final standard-cell instance area; timing and detailed wirelength are post-route. Achieved frequency retains negative slack and is calculated as

$$f_{\max} = \frac{1000}{T_{\text{target}} - \text{WNS}} \text{ MHz}. \quad (1)$$

Routing completion does not imply target feasibility: a zero-DRC result with negative setup slack remains a routed but timing-infeasible point.

### D. Equal-Work Power and Energy

Power uses post-route OpenSTA with deterministic VCD activity. Each timing-feasible implementation receives the same no-error workload of 100,000 accepted 64-bit transactions at II=1 and a 10 ns clock. Both traces contain six reset, 100,000 valid-work, and four drain cycles; four drain cycles exceed the three-cycle pipeline boundary. Input annotation covers the same 139 interface pins before internal propagation. Energy per useful operation is

$$E_{\text{op}} = P T_{\text{trace}} / N_{\text{useful}}. \quad (2)$$

Equal duration and operation count make the power and energy percentage changes identical. This is a steady-stream comparison, not a claim of lower request latency.

The activity experiment also preserves the implementation boundary. Identical input-pin annotation is propagated into each design, so reported totals include the distinct internal switching induced by its RTL and placed-and-routed netlist.

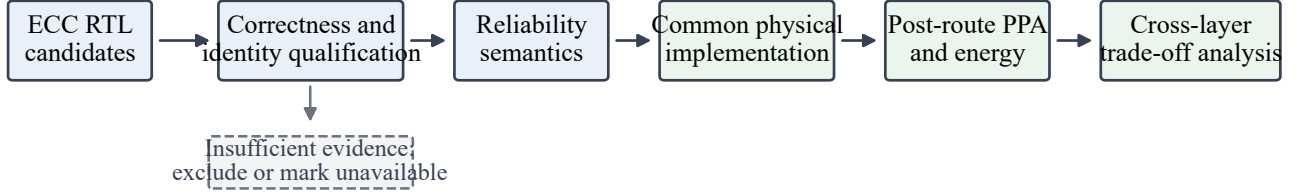


Fig. 1. Identity-preserving evaluation workflow. Insufficient evidence exits the comparison or remains explicitly unavailable; it is never imputed.

We do not decompose the measured difference into register, clock-tree, or combinational contributions; doing so would require an additional experiment. The result therefore supports an empirical equal-work energy contrast, not a causal claim that pipelining generally reduces ECC energy.

#### E. Decision Rule

We avoid a weighted scalar score because its weights could conceal categorical differences in correction guarantee and timing feasibility. Comparisons instead use exact pairwise enumeration. Within one guarantee tier, design  $a$  dominates  $b$  only if it is no worse in every admitted objective and strictly better in at least one. The timing-feasible SECDED objective set minimizes area, latency, and achievable energy per operation while maximizing Fmax. Detailed wirelength is reported as a routing-cost diagnostic and included in sensitivity checks.

Correction guarantees form separate strata: W1 correction/W2 detection and W1/W2 correction are not collapsed into a numeric benefit. A stronger guarantee is therefore not declared globally superior when it is physically worse, and a physically dominating SECDED design does not erase BCH’s stronger guarantee. Timing-infeasible designs retain routed area, wirelength, slack, and achieved Fmax, but they cannot contribute target-clock energy. This rule makes the boundary between measured evidence and design preference explicit.

### IV. EXPERIMENTAL SETUP

All designs protect a 64-bit useful payload. Conventional and Hsiao SECDED use 72-bit codewords; BCH uses 78 bits. The ORFS/OpenROAD toolchain is container-pinned. The platform is SKY130HD at the TT, 1.80 V, 25 °C corner. The common target is 10 ns (100 MHz), with input and output delays each set to 10% of the period, a 0.05 pF output load, 35% core-utilization policy, and 0.55 placement density. Physical runs use deterministic seed 11 and one worker. Encoder and decoder channels process each accepted transaction in parallel.

The common constraint is a reference point, not a promise that every design will meet it. Each implementation receives the same flow stages and policy; no candidate-specific clock relaxation, floorplan tuning, utilization adjustment, or post-result retry is used. A prospective run is retained even when slack is negative. This matters for BCH: relaxing its clock after the 10 ns miss would answer a different question and

would make its power result incomparable with the target-clock SECDED measurements.

Hsiao is correctness/reliability-qualified, but its inferred  $256 \times 73$  table exceeded the unchanged 4096-bit ORFS synthesis-memory policy. The policy was not relaxed after observing the outcome, so Hsiao remains “PPA unavailable” and is excluded from physical conclusions. This status does not imply physical inferiority.

The reported experiment consists of one qualified physical policy and a limited set of implementations, not a seed distribution, process-variation study, multi-corner signoff, or silicon measurement. The artifact package records the pinned image/tool revisions, accepted identities, raw sources, figure transformations, and numerical checks needed to reproduce the manuscript tables and plots.

Post-route area is read from the final design database, timing from the common SDC analysis, and wirelength from detailed routing. Parasitics and activity are propagated for power characterization. The artifact preserves machine-readable implementation identifiers and evidence status beside every value, rather than relying on display labels. Qualification repeats under the fixed seed showed no observed metric variation, but we make no general determinism claim for other hosts, worker counts, or tool revisions.

The manuscript itself is generated from a frozen claim registry. Tables, plots, and numeric macros share the same authoritative evidence source; a semantic validator independently checks the headline quantities and prohibited interpretations. Thus, displayed rounding can be traced to full-precision values without hand-copying numbers between prose and figures.

### V. RESULTS AND ANALYSIS

#### A. Exact-Equivalent SECDED Microarchitecture

Table II and Fig. 2 show that the SECDED architectures occupy distinct physical points despite identical coding semantics. Combinational SECDED uses  $18644.1 \mu\text{m}^2$ ,  $44609 \mu\text{m}$  of detailed wirelength, and reaches 243.881 MHz. Pipelined SECDED uses  $25486.9 \mu\text{m}^2$ ,  $49992 \mu\text{m}$ , and reaches 355.849 MHz. Relative changes are +36.702% area, +12.067% wirelength, and +45.911% Fmax.

The pipeline boundary changes nominal request latency from one to three cycles, while II remains one. It therefore exchanges area and latency for frequency rather than improving all objectives. On the primary objective set (minimize area,

TABLE I  
EVALUATED IMPLEMENTATION IDENTITIES. II IS INITIATION INTERVAL.

| Architecture | Code        | Protection guarantee  | Latency | II | Payload | Physical status      |
|--------------|-------------|-----------------------|---------|----|---------|----------------------|
| SECDED comb. | (72,64)     | W1 correct; W2 detect | 1       | 1  | 64 b    | Routed               |
| SECDED pipe. | (72,64)     | W1 correct; W2 detect | 3       | 1  | 64 b    | Routed               |
| Hsiao SECDED | (72,64)     | W1 correct; W2 detect | 1       | 1  | 64 b    | PPA unavailable      |
| BCH          | (78,64,t=2) | W1/W2 correct         | 1       | 1  | 64 b    | Routed; 10 ns missed |

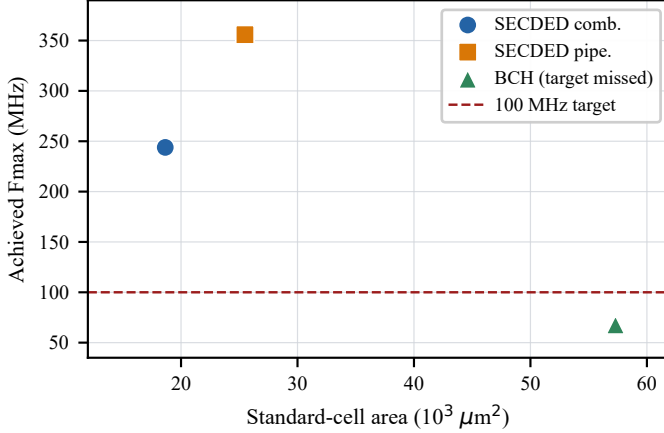


Fig. 2. Post-route area versus achieved Fmax. BCH is routed but misses the 100 MHz target; Hsiao has no artificial point.

latency, and energy; maximize Fmax), neither SECDED point dominates the other.

Cell count provides a consistent secondary check: the combinational and pipelined implementations contain 2,074 and 2,660 standard-cell instances, respectively, a 28.25% increase. Because both implement the same code and transaction semantics, this controlled pair isolates an architecture-sensitive effect that a code-level redundancy or gate-count model would miss. It does not isolate the cost of any one register boundary; placement, buffering, clocking, and routing are jointly reflected in the final netlist.

The equal-work activity study yields 14.575 mW and 145.763 pJ/op for the combinational design, versus 11.075 mW and 110.756 pJ/op for the pipelined design. The latter is a 24.016% reduction in steady-stream power and energy per useful operation. Lower energy and higher request latency are not contradictory: energy integrates activity over equal useful work, whereas latency counts cycles from one request to its response. Reset/drain overhead is present in both traces, and the pipeline sustains one accepted operation per cycle.

This contrast is intentionally evaluated at the shared 100 MHz workload rather than at each design's maximum frequency. It therefore answers which accepted implementation consumes less estimated energy for the same amount of useful work under the common feasible operating point. It does not project voltage-frequency scaling, idle behavior, bursty traffic, or energy at the reported Fmax values.

### B. Correction Capability versus Physical Feasibility

The evaluated BCH (78,64,t = 2) RTL raises the proven guarantee from W1 correction/W2 detection to W1/W2 correction. That stronger guarantee coexists with substantial implementation cost: relative to combinational SECDED, standard-cell area rises by 207.356%, detailed wirelength by 310.975%, and achieved Fmax falls by 72.258%.

BCH completes routing with zero final design-rule violations, but has -4.78052 ns WNS at the 10 ns constraint. Equation (1) gives an achieved Fmax of 67.6566 MHz, below the common 100 MHz target. Its target-constraint power remains a diagnostic analysis value; target-clock energy is excluded because 100 MHz is not an achievable operating point for this implementation. The result characterizes the evaluated syndrome/Chien RTL under the stated physical environment. It is not a family-wide frequency or cost bound on BCH implementations.

The physical spread is not a small perturbation. BCH contains 7,129 cells, 243.73% more than combinational SECDED, while its six additional codeword bits account for only part of the realized decoder structure. The result therefore motivates architecture-aware exploration, but it cannot distinguish unavoidable algebraic complexity from avoidable characteristics of this RTL. That distinction is why the implementation identifier accompanies every value and why the claim remains scoped to this syndrome/Chien realization.

### C. Design-Space Interpretation

Figure 3 separates all routed physical metrics from timing-feasible energy. BCH is retained in the routed panel, including its lower achieved Fmax, but omitted from energy. Hsiao is omitted rather than placed at an artificial coordinate. This separation prevents a timing-infeasible power estimate from masquerading as an operating point.

Exact enumeration confirms two different readings of the design space. Within the timing-feasible SECDED tier, neither microarchitecture dominates: combinational wins area and latency, whereas pipelined wins Fmax and achievable energy. In a reliability-agnostic routed-physical space, combinational SECDED dominates the evaluated BCH point because it has lower area, lower cell count, lower wirelength, no target deficit, equal one-cycle latency, and higher Fmax. Excluding latency does not change that sensitivity result; both SECDED implementations dominate BCH on the remaining physical objectives.

These physical dominance statements are intentionally not global ECC rankings. BCH occupies a stronger guarantee stratum, so its W2 correction cannot be exchanged for area by

TABLE II  
POST-ROUTE RESULTS. POWER AND ENERGY ARE COMPARABLE ONLY FOR THE TWO TIMING-FEASIBLE SECDED POINTS.

| Architecture | Area ( $\mu\text{m}^2$ ) | Fmax (MHz) | Wirelength ( $\mu\text{m}$ ) | Power (mW) | Energy (pJ/op) | 10 ns           |
|--------------|--------------------------|------------|------------------------------|------------|----------------|-----------------|
| SECDED comb. | 18644.1                  | 243.881    | 44609                        | 14.575     | 145.763        | met             |
| SECDED pipe. | 25486.9                  | 355.849    | 49992                        | 11.075     | 110.756        | met             |
| Hsiao SECDED | —                        | —          | —                            | —          | —              | PPA unavailable |
| BCH          | 57303.7                  | 67.6566    | 183332                       | diagnostic | —              | missed          |

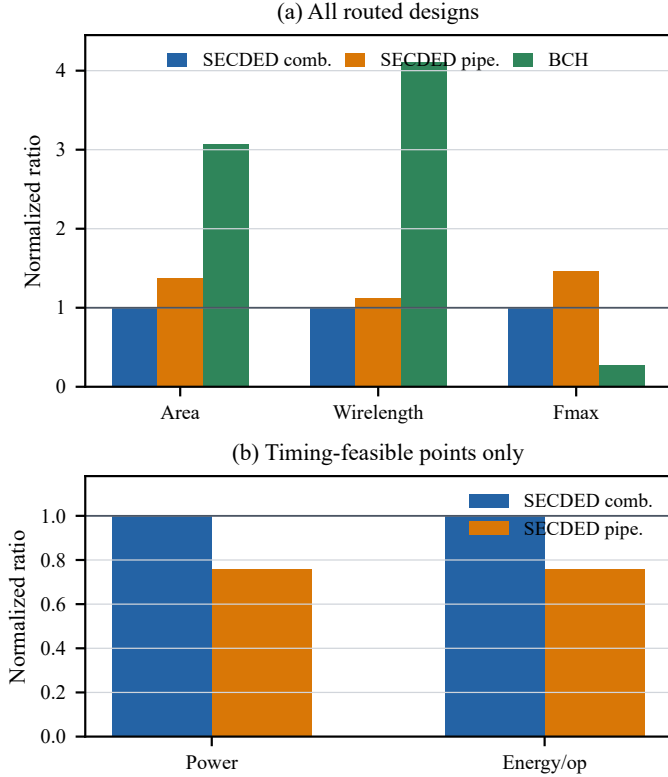


Fig. 3. Normalized trade-offs. (a) Area, detailed wirelength, and achieved Fmax for all routed designs. (b) No-error power and energy/op only for timing-feasible designs. Ratios use combinational SECDED as 1.0.

an unstated preference weight. A system requiring correction of every double-coordinate error may reject both SECDED points regardless of physical cost; a system requiring 100 MHz under this exact implementation set cannot select BCH. The method exposes that feasibility boundary without pretending to resolve the system-level requirement.

The exhaustive weight-3 observations provide secondary context. Conventional SECDED produces 45,304 SDC and 14,336 DUE outcomes among 59,640 masks; Hsiao produces 34,164 SDC and 25,476 DUE outcomes over the same universe; BCH produces 13,780 SDC and 62,296 DUE outcomes among 76,076 masks. Reduced fractions are respectively 809/1065 and 256/1065, 2847/4970 and 2123/4970, and 265/1463 and 1198/1463. Because coordinate universes differ and no field weighting is applied, these counts do not rank operational reliability.

Even the same-size SECDED universe requires care: Hsiao's

lower W3 SDC count is an exact canonical-coordinate property of the evaluated parity-check construction, not an operational failure rate. Workload data values, physical bit adjacency, upset multiplicity, interleaving, and scrubbing can change which masks matter in a deployed memory. The counts are useful for checking decoder behavior beyond its guarantee and for exposing structural differences, but not for assigning expected field failures.

## VI. LIMITATIONS AND DISCUSSION

**Code semantics are insufficient.** The controlled SECDED pair shows that exact-equivalent coding behavior does not uniquely determine implementation behavior. Register placement changes area, interconnect, latency, Fmax, and activity, creating a non-dominated trade rather than one universally better implementation.

**Stronger correction carries implementation cost.** The BCH point corrects all double-coordinate errors that SECDED only detects, but its evaluated realization is larger, more routing-intensive, and infeasible at the shared target. The comparison intentionally retains the extra six codeword bits as part of the realized cost, so it cannot attribute every difference solely to decoder algebra. Different BCH architectures could occupy different points [9], [10].

**Missing data must remain missing.** Hsiao illustrates why qualification boundaries matter. Relaxing the synthesis-memory policy only after observing failure would create a different experiment; imputing PPA would be less defensible still. Explicit missingness narrows the conclusion without suggesting that Hsiao is physically inferior.

The empirical scope is limited to specific RTL implementations, SKY130HD, one TT 1.8 V/25 °C corner, one common 10 ns target, and one deterministic seed/worker policy. Only four reliability identities, three routed designs, and two timing-feasible energy points are available. OpenSTA activity-based power is an estimate, not a silicon measurement. BCH target-clock energy is excluded, and Hsiao PPA is unavailable. We do not study process or seed distributions, multi-corner signoff, physical SRAM arrays, bit mapping, interleaving, scrubbing, or memory-macro cost. No FIT, SER, field weighting, or operational SDC/DUE probability is established. These limits bound generalization while leaving the controlled within-flow contrasts intact.

### A. Threats to Validity

**Internal validity.** A shared policy reduces configuration confounding but does not remove heuristic implementation

effects. One seed and one worker provide a controlled point, not a distribution over place-and-route outcomes. The prospective no-retry rule prevents selectively polishing a poor result, while also leaving open whether a different equally fair policy would alter the ordering. Exact SECEDED equivalence and the common transaction contract strengthen the microarchitecture contrast; the BCH comparison lacks such an exact-equivalent architectural pair.

*Construct validity.* Final cell area, routed wirelength, post-route timing, and activity-based power operationalize implementation cost, but do not cover every system cost. The experiment excludes SRAM macro area, ECC storage organization, clock-domain crossings, control integration, and error-recovery latency. The equal-work trace supports a throughput-mode energy comparison, not energy under arbitrary workloads. Similarly, canonical-coordinate W3 enumeration measures decoder outcomes in a finite logical universe, not radiation susceptibility.

*External validity.* A result from one open 130-nm library and one corner need not transfer quantitatively to advanced nodes, other cell libraries, memories, or decoder generators. The conclusion with the broadest support is therefore methodological: preserve RTL identity, guarantees, feasibility, and missingness when joining reliability with physical evidence. The numerical trade-offs remain bounded to the named artifacts.

### B. Implications for ECC Selection

The evidence suggests a two-stage decision rather than a single ranking. First, establish non-negotiable requirements: correction guarantee, target clock, transaction latency, and acceptable evidence status. Second, compare surviving implementation identities on physical objectives. In this experiment, a strict W2-correction requirement admits BCH semantically but exposes a timing implementation gap; a strict 100 MHz requirement admits the two SECEDED points but not BCH; and a three-cycle latency prohibition favors combinational SECEDED despite the pipeline’s energy and Fmax advantages. Reporting these conditional choices is more actionable than naming one “best ECC.”

For designers, the missing Hsiao point is also informative. It identifies a representation boundary—an inferred table under a fixed synthesis-memory policy—that future work could address prospectively with an algorithmic encoder/decoder RTL. Such a future implementation would be a new identity and should be requalified, rather than used retroactively to fill the present table.

### C. Auditability and Failure-Closed Reporting

Cross-layer studies are vulnerable to accidental evidence substitution: a proof from one RTL can be joined to PPA from another, a pre-route estimate can be displayed beside post-route measurements, or a power report can survive after its clock point becomes infeasible. Our artifact guards the joins with stable implementation identifiers and evidence status. A physical row is admitted only when its identity matches the qualified RTL and the common flow policy. The same rule

propagates to derived quantities: Fmax retains measured WNS, and target-clock energy exists only when that target is feasible.

The presentation has three additional safeguards. First, guarantee statements and exhaustive W3 observations occupy different fields and are described with different language. Second, unavailable values remain textual status, not zeros, infinities, or coordinates in a plot. Third, full-precision evidence feeds generated macros, CSV tables, and plots, while manuscript rounding is display-only. The reported 36.702%, 45.911%, and 24.016% SECEDED changes, for example, are recomputed from the frozen raw values rather than copied from captions.

This discipline changes what a reviewer can falsify. The SECEDED equivalence claim can be checked independently of physical cost; the BCH timing status can be recomputed from target period and WNS; the power comparison can be audited against operation count, trace duration, II, and feasibility; and every omitted point has a stated reason. None of these checks proves external validity, but they make the implementation-scoped claims inspectable and prevent an unsupported value from silently strengthening the result.

### D. What Would Change the Decision

The present Pareto structure is contingent rather than immutable. A prospectively defined Hsiao RTL that passes the unchanged qualification and physical policy could add a new point without altering the current measurements. An alternative BCH architecture could meet 10 ns and thereby become eligible for equal-work target-clock energy, but its result could not be substituted for the evaluated implementation. Multi-seed or multi-corner characterization could reveal distributions around each point, while a physically grounded fault model could supply weights for the W3 masks. Each extension answers a distinct uncertainty and should preserve the same identity and feasibility rules.

These extensions also clarify which conclusions are already stable. No future point can make the two measured SECEDED implementations code-inequivalent, erase their observed area/Fmax/energy trade-off under this environment, or turn the measured BCH WNS into a met constraint. Future evidence can broaden the design space and alter a selection, but should not rewrite the status of the frozen points.

## VII. CONCLUSION

ECC selection should preserve implementation identity rather than rely on coding capability alone. Under one pinned post-route flow, exact-equivalent combinational and pipelined SECEDED architectures exhibit distinct, non-dominated physical trade-offs, while the stronger-correcting evaluated BCH RTL incurs substantial routed cost and misses the common timing target. A reproducible comparison can expose these choices without treating unavailable or infeasible evidence as favorable measurements. Broader conclusions require more technologies, implementations, corners, and physically grounded fault models.

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